

Week 4 - Lesson 1 - Hypotenuse

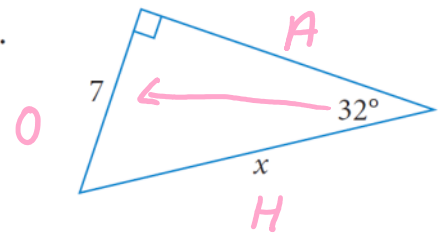
Thursday, 9 April 2020 10:57 AM

Example 1

Calculate the length of the hypotenuse, correct to 1 decimal place.

SOH CAH TOA

$$\sin \theta = \frac{O}{H}$$



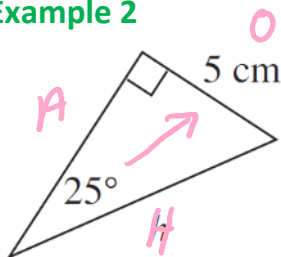
$$x \times \sin 32^\circ = \frac{7}{x} \times x$$

$$x = \frac{7}{\sin 32^\circ}$$

$$x \times \frac{\sin 32^\circ}{\sin 32^\circ} = \frac{7}{\sin 32^\circ}$$

$$x = 13.2$$

Example 2



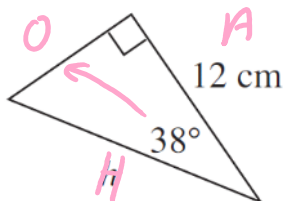
SOH CAH TOA

$$H \times \sin 25^\circ = \frac{5}{H} \times H$$

$$H \times \frac{\sin 25^\circ}{\sin 25^\circ} = \frac{5}{\sin 25^\circ}$$

$$H = \frac{5}{\sin 25^\circ} = \underline{\underline{11.8 \text{ cm}}}$$

Example 3



SOH CAH TOA

$$H \times \cos 38^\circ = \frac{12}{H} \times H$$

$$H \times \frac{\cos 38^\circ}{\cos 38^\circ} = \frac{12}{\cos 38^\circ}$$

$$H = \frac{12}{\cos 38^\circ} = 15.2 \text{ cm}$$

